

Vaibhavi Jaiswal

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EDUCATION

Vellore Institute of Technology (VIT)

Chennai, India

Bachelor of Technology in Artificial Intelligence and Machine Learning (CGPA: 8.38) July 2023 – July 2027 (Expected)

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

AI & Deep Learning: LangGraph, LangChain, PyTorch, TensorFlow, Scikit-Learn, Gemini API, ChromaDB, FAISS

Frameworks & Tools: React, Node.js, Express.js, Flask, Streamlit, Docker, Git, GitHub Actions, Vercel

Databases & Stores: MySQL, MongoDB, PostgreSQL, Redis, Vector Stores

PROJECTS & RESEARCH

Cyclone Tracking & Intensity Prediction using Vision Transformers | *PyTorch, Python* Research Pipeline

- Co-architected an optimized Vision Transformer (ViT) configuration in a cross-functional team, evaluating spectral satellite telemetry matrices to minimize spatial variant error anomalies.
- Collaborated on the training and tuning loops to achieve a structural vector intensity prediction accuracy of **93% R^2 score** and an MAE of **5.54** during rigorous neural validation testing.

Autonomous AI Code Optimization Agent | *LangGraph, Docker, Gemini API, FAISS, Python*

- Engineered a production multi-agent system utilizing LangGraph to build iterative code profiling and runtime optimization graphs.
- Isolated code execution flows securely inside dynamically managed Docker sandboxes to map structural bottlenecks safely.
- Integrated an automated semantic loop that cross-checks code targets against indexed software engineering design patterns to generate valid algorithmic optimizations verified by unit boundaries.

Modern School Hub Platform (SRJIC) | *React, Tailwind CSS, Vercel, Framer Motion* Freelance / Production

- Engineered a responsive, SEO-optimized educational platform featuring decoupled interaction loops to ensure a **0ms Total Blocking Time (TBT)** on the main thread.
- Deployed a highly optimized asset rendering architecture via global edge middleware routing paths, resulting in a **0.70s First Contentful Paint (FCP)**.
- Achieved a validated **98/100 production Lighthouse Performance Index** for discoverability, access speeds, and mobile-first micro-interactions.

AI Repository Assistant | *LangChain, ChromaDB, Gemini AI, HuggingFace, Streamlit*

- Developed a RAG-powered interactive codebase parser using language-aware Abstract Syntax Tree (AST) semantic chunking to capture deep technical contexts across complex external architectures.
- Constructed an optimized retrieval matrix using Maximal Marginal Relevance (MMR) over vector space representations hosted natively in ChromaDB indexes.

ACHIEVEMENTS

- 3rd Place Winner:** Ranked 3rd out of hundreds of peer competitors in a highly competitive coding battle among 3rd-year students at VIT Chennai.
- Hackathon Finalist:** Secured a spot as a **Top 20 Finalist** at Hack Night '25, an elite inter-college hackathon championship.

CERTIFICATIONS & EXTRACURRICULARS

IBM Watsonx AI Assistant Certified Specialist – issued by IBM

Full-Stack Web Development Credential – issued via Udemy